Matthew Cho

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EDUCATION

McGill University (GPA: 4.0/4.0) | M. Sc. in Biological and Biomedical Engineering 2023 – December 2025
McGill University (GPA: 3.8/4.0) | B.Eng. in Bioengineering, graduated with distinction. 2018 – May 2022

SKILLS

Programming: Python, SQL, Java, MATLAB, JavaScript, HTML/CSS, C, Git.
Data Science: Pandas, NumPy, scikit-learn, PyTorch, statistical analysis, predictive modeling, visualization, machine learning
Data Engineering: PostgreSQL, MySQL, SQLAlchemy, FastAPI, ETL/data pipelines, database design, data ingestion
Laboratory: Cell culture, PCR, electrophoresis, optical microscopy, animal handling and husbandry, technical writing.

PUBLICATIONS

Cho, M. (2026). *ProMS: An Integrated Computational Framework for Molecular Surface Analysis and Atomic-Level Hydrophobicity Mapping* (Master's thesis). McGill University.

Perumal, F. A. S., **Cho**, **M.**, et al. (2026). Amplification of computational power by the multiplication of bacteria exploring microfluidic networks encoding mathematical problems. *Nature Microsystems & Nanoengineering*.

Cho, M., et al. (2024). Protein adsorption on solid surfaces: Data mining, database, molecular surface-derived properties, and semiempirical relationships. *ACS Applied Materials & Interfaces*, 16(22), 28290–28306.

EXPERIENCE

Research Assistant | McGill University May 2021 – December 2022, December 2025 - Current
- Developed computational models and software for studying molecular surfaces and biological-agent behaviour.
- Designed data-processing and analysis workflows for complex biological datasets, built custom visualization modes.
- Built reproducible research tools and contributed significant computational analyses to peer-reviewed publications.
- Supervised and provided guidance to several undergraduate researchers working on computational biology projects.

Teaching Assistant / Course Assistant | McGill University Fall 2024, Winter 2026
- Served as a TA for BIEN540 (2026) and course assistant for BIEN510 (2024), two graduate-level courses.
- Designed and led weekly two-hour tutorials covering course concepts, exam preparation, and project development.
- Graded coursework, answered student questions, and provided technical and conceptual guidance to 40+ students.

PROJECTS

Protein Molecular Surface Calculator (ProMS) | Python, C, NumPy, Pandas, PySide/Qt, PyMOL
- Developed an object-oriented software package for quantitative analysis and 3D visualization of molecular surfaces.
- Built a reproducible pipeline for PDB processing, molecular-surface generation, atomic property mapping, aggregation, and interactive PyMOL visualization of proteins, nucleic acids, and small molecules.
- Designed a Qt graphical interface and batch-processing workflow and deployed it as a PyMOL standalone plugin.
- Distributed the tool to researchers and collaborators across multiple universities and biotechnology organizations.

Edgework Analytics | Python, PostgreSQL, Pandas, scikit-learn, SQLAlchemy, FastAPI, HTML/CSS, JavaScript
- Developing an automated data and machine-learning platform for MLB performance analysis and betting evaluation.
- Built ETL pipelines integrating schedules, box scores, Statcast data, player statistics, rosters, lineups, and sportsbook odds into PostgreSQL. Designed historical and point-in-time data models to support reproducible model training.
- Developed predictive models using multi-window performance features and probability calibration, with model probabilities evaluated against market prices using expected value. Performed thorough backtesting without leakage.
- Built automated prediction archiving and an interactive web dashboard for the easy monitoring of model outputs.

Computational Simulations with Biological Agents Suite | Python, NumPy, OpenCV, Java, C, MATLAB
- Developed agent-based growth models of bacterial and fungal behaviour in complex microfluidic environments.
- Applied image processing and computer vision to convert experimental maze geometries into computational simulation environments. Built interactive visualization and GUI tools to perform repeated simulations.